JULKAR NIENE

Software Engineer · Full Stack & AI Systems · React.js/ Next.js/ Python / FastAPI / Node.js

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PROFESSIONAL SUMMARY

CSE graduate with a Data Science major who builds backend systems, APIs, and AI-integrated tools end-to-end. Primary stack is Python/FastAPI and Node.js/Express, with hands-on experience in PostgreSQL schema design, REST API architecture, JWT auth, real-time systems, and LLM integration. Comfortable entering unclear problems, tracing issues to root cause, and shipping working software without needing perfect instructions. Uses AI development tools (Claude, Cursor, Copilot) to prototype faster and write sharper code without blindly trusting the output. Strong written communicator: Swagger/OpenAPI docs, technical READMEs, and system specs.

TECHNICAL SKILLS

Languages	Python, JavaScript (ES6+), SQL, C++, HTML5/CSS3
Backend	FastAPI, Node.js/Express, REST API design, JWT auth, Socket.io, Prisma ORM
Databases	PostgreSQL, MongoDB, MySQL
AI & ML	LLM API integration, RAG pipelines, LangChain basics, LoRA fine-tuning, NLP/text classification
AI Dev Tools	Claude Code, Cursor, GitHub Copilot
Frontend	React.js, Next.js 14, Tailwind CSS, Zustand, Chart.js
Tooling	Git/GitHub, Swagger/OpenAPI, Postman, Docker (basics), Railway, Cloudinary, Jira, Figma
Practices	System design, RBAC, Monorepo architecture, Agile/Scrum, technical documentation

PROJECTS

Collaborative Team Hub

Stack: *Next.js 14, Node.js/Express, PostgreSQL/Prisma, Socket.io, Zustand, Cloudinary, Railway*

- Architected and built a production-ready monorepo from scratch , backend API (Node.js/Express), frontend (Next.js 14), and PostgreSQL database with Prisma ORM, all deployed on Railway under assessment deadline.
- Designed RBAC (role-based access control) schema and implemented JWT auth; built real-time messaging with Socket.io and optimistic UI updates for low-latency UX.
- Wrote full Swagger/OpenAPI documentation for all endpoints; integrated Cloudinary for media uploads and Resend for transactional email shipped as a complete, tested system.
- Debugged schema migration edge cases and race conditions in real-time event handling — traced issues to root cause and documented fixes.

LifeOS : Personal AI Productivity Backend

Stack: *Python (FastAPI), PostgreSQL, Groq AI API, JWT, REST*

- Designed and built a multi-module REST API (task management, expense tracking, AI querying) with FastAPI and PostgreSQL including relational schema design, CRUD, JWT auth, and background processing.
- Integrated Groq LLM API for natural language querying over personal data (e.g., 'summarise my spending this month') handling prompt construction, response parsing, and structured output extraction.
- Built dynamic data aggregation endpoints feeding Chart.js visualisations; iterated schema and API contracts based on real usage.

AI Job Application Tracking System

Stack: *React.js, FastAPI, Supabase/PostgreSQL, Groq LLM API, PDF/DOCX parsing*

- Built an automation pipeline that parses raw job descriptions via LLM API, extracts structured fields (skills, experience, salary range), and stores results in a normalised PostgreSQL schema.
- Implemented CV parsing (PDF/DOCX ingestion), semantic matching logic, and auto-generated cover letter generation — end-to-end AI workflow with real data.
- Handled edge cases in document parsing (malformed PDFs, inconsistent JD formats) and iterated on prompt engineering to improve extraction accuracy.

LWNR-LoRA — Parameter-Efficient LLM Fine-Tuning (Final Year Project)

Stack: *Python, PyTorch, HuggingFace Transformers, LoRA*

- Researched and implemented LoRA-based fine-tuning achieving 54.2% parameter reduction and 2.38× inference speedup on BERT Base, measured and benchmarked against baseline.
- Produced research-grade documentation: benchmark tables, ablation results, BibTeX citation, and a full reproducibility guide (notebook cell walkthrough).

WORK EXPERIENCE

Co-Head of HR — UIU App Forum (University Tech Club) *Sep 2024 – Oct 2025*

- Led recruitment cycles for a 100+ member tech club: defined role requirements, screened candidates, coordinated interviews, and onboarded selected members.
- Managed cross-functional coordination across development, design, and events teams; developed clear written communication habits for async team updates.

Grader — United International University *Jun 2023 – Oct 2023*

- Evaluated programming assignments and exam scripts for correctness, code quality, and algorithmic logic across multiple course cohorts.
- Provided structured written feedback to students; maintained consistent grading standards in collaboration with faculty.

EDUCATION

B.Sc in Computer Science & Engineering — United International University (UIU) *2022 – 2026 | Graduated*

Major: Data Science | CGPA: 3.4 / 4.00

ACHIEVEMENTS & CERTIFICATIONS

- LWNR-LoRA (Final Year Project): 54.2% parameter reduction and 2.38× inference speedup on BERT Base — research-grade documentation with benchmark tables and BibTeX.
- 3rd Runner Up — UIU CSE Project Show (Database Management System)
- CCNA: Introduction to Networks — Cisco Certified